

Problem	Language	Time	Result	Score	Action
Roads and Libraries	cpp	25 minutes	Accepted	100.0	View results
Queues: A Tale of Two Stacks	cpp	about 1 hour	Accepted	30.0	View results
Queues: A Tale of Two Stacks	cpp	about 1 hour	Accepted	30.0	View results
Cycle Detection	cpp	about 1 hour	Accepted	5.0	View results
Castle on the Grid	cpp	about 1 hour	Accepted	30.0	View results
Min Max Riddle	cpp	about 1 hour	Accepted	60.0	View results
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Problem	Language	Time	Result	Score	Action
Largest Rectangle	cpp20	about 2 hours	Accepted	50.0	View results
Balanced Brackets	cpp20	2 days	Accepted	25.0	View results

Largest Rectangle

```
13  * THE FUNCTION ACCEPTS INTEGER_ARRAY h AS PARAMETER.
14  */
15
16  ✓ long largestRectangle(vector<int> h) {
17      stack<int> s;
18      long maxArea=0;
19      int n=h.size();
20      int i=0;
21  ✓ while (i<n){
22      ✓ if (s.empty() || h[s.top()] <= h[i])
23          s.push(i++);
24  ✓ else {
25          int tp = s.top();
26          s.pop();
27          long area_top = (long)h[tp] * (s.empty() ? i : i - s.top() - 1);
28          maxArea = max(maxArea, area_top);
29      }
30  }
31  ✓ while (!s.empty()) {
32      int tp = s.top();
33      s.pop();
34      long area_top = (long)h[tp] * (s.empty() ? i : i - s.top() - 1);
35      maxArea = max(maxArea, area_top);
36  }
37
38      return maxArea;
39  }
40
41
```

✓ Test case 0

Compiler Message

✓ Test case 1

Success

✓ Test case 2

Input (stdin)

Download

```
1 5
2 1 2 3 4 5
```

✓ Test case 3

✓ Test case 4

Expected Output

Download

```
1 9
```

✓ Test case 5

✓ Test case 6

Roads and Libraries

```
--
20 ✓ long roadsAndLibraries(int n, int c_lib, int c_road, vector<vector<int>> cities) {
21 ✓ if (c_lib <= c_road) return (long)n * c_lib;
22
23     vector<vector<int>> adj(n + 1);
24 ✓ for (auto &road : cities) {
25     adj[road[0]].push_back(road[1]);
26     adj[road[1]].push_back(road[0]);
27 }
28
29 vector<bool> visited(n + 1, false);
30 long totalCost = 0;
31
32 ✓ for (int i = 1; i <= n; i++) {
33 ✓     if (!visited[i]) {
34         queue<int> q;
35         q.push(i);
36         visited[i] = true;
37         int size = 0;
38
39         while (!q.empty()) {
40             int u = q.front(); q.pop();
41             size++;
42             for (int v : adj[u]) {
43                 if (!visited[v]) {
44                     visited[v] = true;
45                     q.push(v);
46                 }
47             }
48         }
49
50         totalCost += c_lib + (long)(size - 1) * c_road;
51     }
52 }
53
54 return totalCost;
55 }
56
57 int main()
58 {
```

```
15 * 2. INTEGER c_lib
17 * 3. INTEGER
```

Test case 0

Compiler Message

Test case 1

Success

Test case 2

Input (stdin)

Download

Test case 3

```
1 2
2 3 3 2 1
3 1 2
```

Test case 4

```
4 3 1
5 2 3
```

Test case 5

```
6 6 6 2 5
7 1 3
```

Test case 6

```
8 3 4
9 2 4
```

Cycle Detection

```
68  * };
69  *
70  */
71  bool has_cycle(SinglyLinkedListNode* head) {
72
73      if (!head) return false;
74
75      SinglyLinkedListNode *slow = head;
76      SinglyLinkedListNode *fast = head;
77
78      while (fast && fast->next) {
79          slow = slow->next;
80          fast = fast->next->next;
81
82          if (slow == fast)
83              return true;
84      }
85
86      return false;
87  }
88
89
```

Test case 0

Compiler Message

Test case 1

Success

Test case 2

Input (stdin)

Download

1 1

Test case 3

2 -1

3 1

Test case 4

4 1

Test case 5

Expected Output

Download

1 0

Test case 6

Min Max Riddle

```
10 vector<long> riddle(vector<long> arr) {
11     int n = arr.size();
12     vector<long> res(n, 0);
13     vector<int> left(n), right(n);
14     stack<int> s;
15     for (int i = 0; i < n; i++) {
16         while (!s.empty() && arr[s.top()] >= arr[i])
17             s.pop();
18         left[i] = s.empty() ? -1 : s.top();
19         s.push(i);
20     }
21
22     while (!s.empty()) s.pop();
23     for (int i = n - 1; i >= 0; i--) {
24         while (!s.empty() && arr[s.top()] >= arr[i])
25             s.pop();
26         right[i] = s.empty() ? n : s.top();
27         s.push(i);
28     }
29     for (int i = 0; i < n; i++) {
30         int len = right[i] - left[i] - 1;
31         res[len - 1] = max(res[len - 1], arr[i]);
32     }
33     for (int i = n - 2; i >= 0; i--) {
34         res[i] = max(res[i], res[i + 1]);
35     }
36
37     return res;
38 }
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)

```
1 4
2 2 6 1 12
```

Download

✔ Sample Test case 1

✔ Sample Test case 2

Your Output (stdout)

```
1 12 2 1 1
```

Expected Output

```
1 12 2 1 1
```

Download

Castle on the Grid

```
21 int minimumMoves(vector<string> grid, int startX, int startY, int goalX, int goalY) {
22     int n = grid.size();
23     vector<vector<bool>> visited(n, vector<bool>(n, false));
24
25     queue<pair<pair<int,int>, int>> q;
26     q.push({startX, startY, 0});
27     visited[startX][startY] = true;
28
29     int dx[] = {1, -1, 0, 0};
30     int dy[] = {0, 0, 1, -1};
31
32     while (!q.empty()) {
33         auto curr = q.front();
34         q.pop();
35
36         int x = curr.first.first;
37         int y = curr.first.second;
38         int moves = curr.second;
39
40         if (x == goalX && y == goalY)
41             return moves;
42         for (int d = 0; d < 4; d++) {
43             int nx = x;
44             int ny = y;
45             while (true) {
46                 nx += dx[d];
47                 ny += dy[d];
48                 if (nx < 0 || ny < 0 || nx >= n || ny >= n || grid[nx][ny] == 'X')
49                     break;
50
51                 if (!visited[nx][ny]) {
52                     visited[nx][ny] = true;
53                     q.push({nx, ny, moves + 1});
54                 }
55             }
56         }
57     }
58     return -1;
59 }
60 }
```

140

Line: 140 Col: 1

 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

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✓ Sample Test case 1

✓ Sample Test case 2

```
1 3
2 .X.
3 .X.
4 ...
5 0 0 0 2
```

Your Output (stdout)

```
1 3
```

Expected Output

```
1 3
```

[Download](#)

Tale of two stacks

```
10 class MyQueue {
11
12 private:
13     stack<int> s1, s2;
14
15 public:
16     void enqueue(int x) {
17         s1.push(x);
18     }
19
20     void shiftStacks() {
21         if (s2.empty()) {
22             while (!s1.empty()) {
23                 s2.push(s1.top());
24                 s1.pop();
25             }
26         }
27     }
28
29     void dequeue() {
30         shiftStacks();
31         s2.pop();
32     }
33
34     int front() {
35         shiftStacks();
36         return s2.top();
37     }
38 };
39
40 int main() {
41     MyQueue q1;
42     int q, type, x;
43     cin >> q;
44
45     for(int i = 0; i < q; i++) {
46         cin >> type;
47         if(type == 1) {
48             cin >> x;
49             q1.enqueue(x);
50         }
51         else cout << q1.front() << endl;
52     }
53     /* Enter your code here. Read input from STDIN. Print output to STDOUT */
54     return 0;
55 }
```

Line: 59 Col: 1

 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

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✔ Sample Test case 0

Input (stdin)

Download

✔ Sample Test case 1

✔ Sample Test case 2

```
1 10
2 1 42
3 2
4 1 14
5 3
6 1 28
7 3
8 1 60
9 1 78
10 2
11 2
```

... < > >> >>>

Balanced Brackets :

```
string isBalanced(string s) {  
    stack<char> st;  
    for(char c : s){  
        if(c=='(' || c=='[' || c=='{') st.push(c);  
        else{  
            if(st.empty()){  
                return "NO";  
            }  
            char top=st.top();  
            st.pop();  
            if(c==')' && top!='(') return "NO";  
            if(c==']' && top!='[') return "NO";  
            if(c=='}' && top!='{') return "NO";  
        }  
    }  
    return st.empty()? "YES" : "NO";  
}
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Sample Test case 2

Input (stdin)

```
1 3  
2 {[()]}  
3 {[()]}  
4 {[[(())]]}
```

Your Output (stdout)

```
1 YES  
2 NO  
3 YES
```

Expected Output

```
1 YES
```

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